WEEK -8

Minor Project (BTCS607N)-Weekly Progress Report

Project Title: Drone-Based Intelligent System for Apple Orchard Management in Himachal Pradesh.

Group ID: G9

Branch & Section: BTech CSE (AIML with Microsoft), AIDS -A

Date: 27-03-2026

1. Executive Summary

A. **Current Sprint:** Sprint 1 (Issue – 8)

B. **Sprint / Issue Goal:** To implement core system components, including dataset preparation, frontend validation, and database creation.

C. **Status:** On Track

   I. Core components have been implemented successfully, and the system is ready for backend integration.

2. UML & OOAD Progress

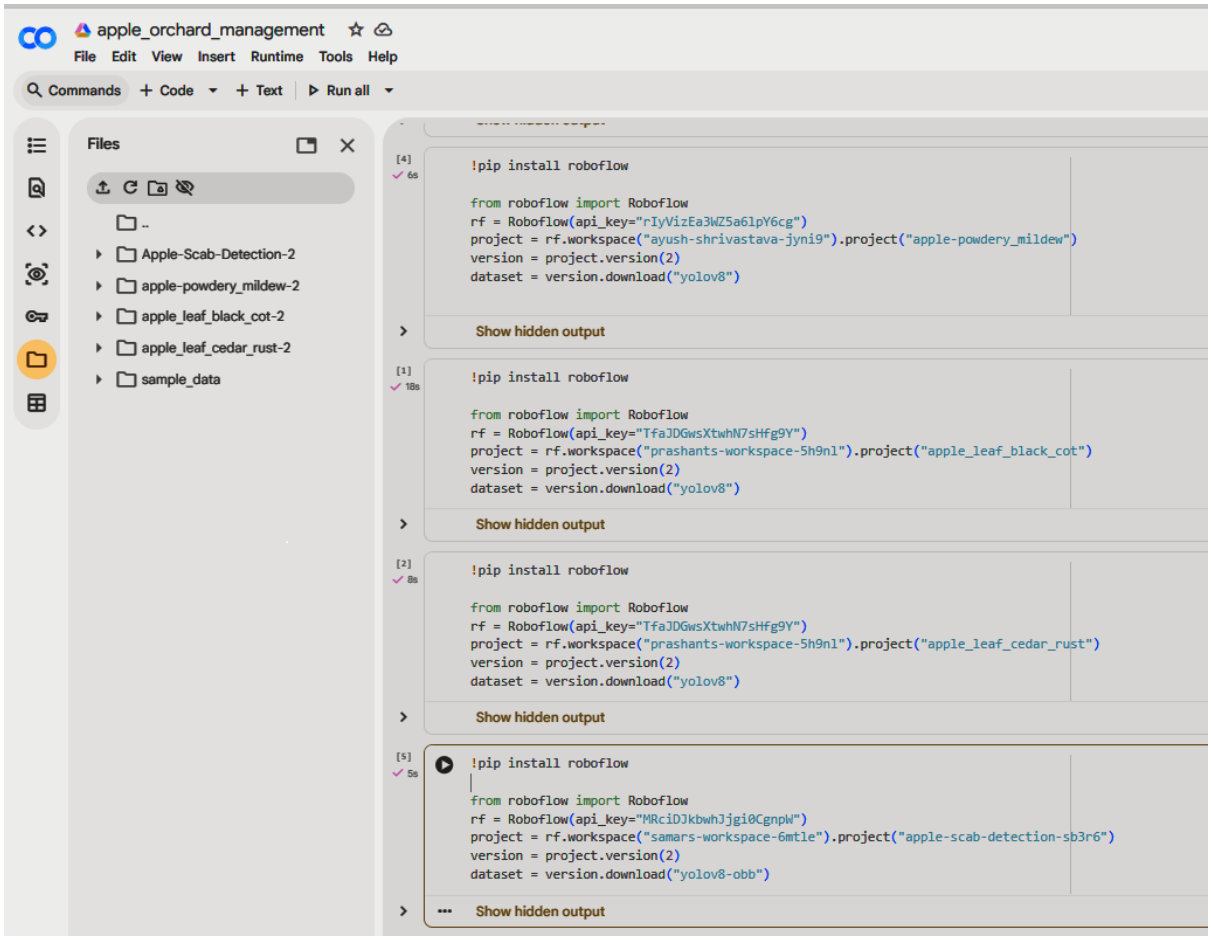
A. **Diagrams Completed this week:** No new diagrams were created this week.

B. **Diagrams Revised:** No revisions were made to existing diagrams.

C. **OO Concept Applied:**

   1. **Encapsulation:** Implemented logical separation of components such as preprocessing, detection, and database handling.
   2. **Abstraction:** Used previously created UML diagrams (Class and Sequence Diagrams) to guide implementation.
   3. **Modularity:** Divided system into independent modules such as frontend, dataset processing, and database.
   4. **Separation of Concerns:** Each component (UI, dataset, database) was handled independently for better maintainability.

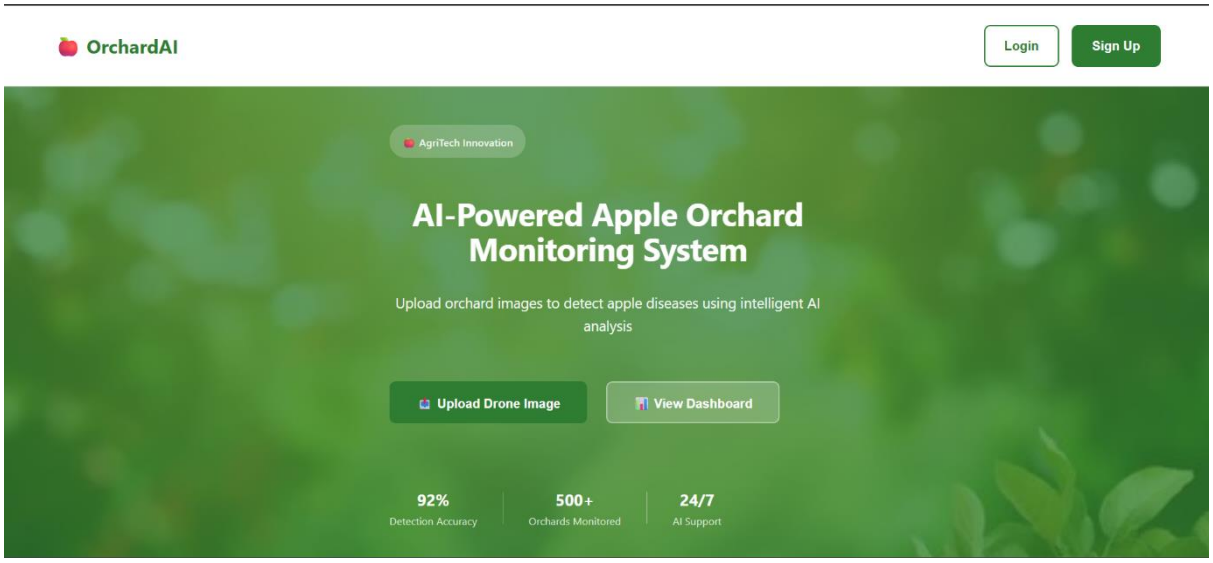
Dataset:

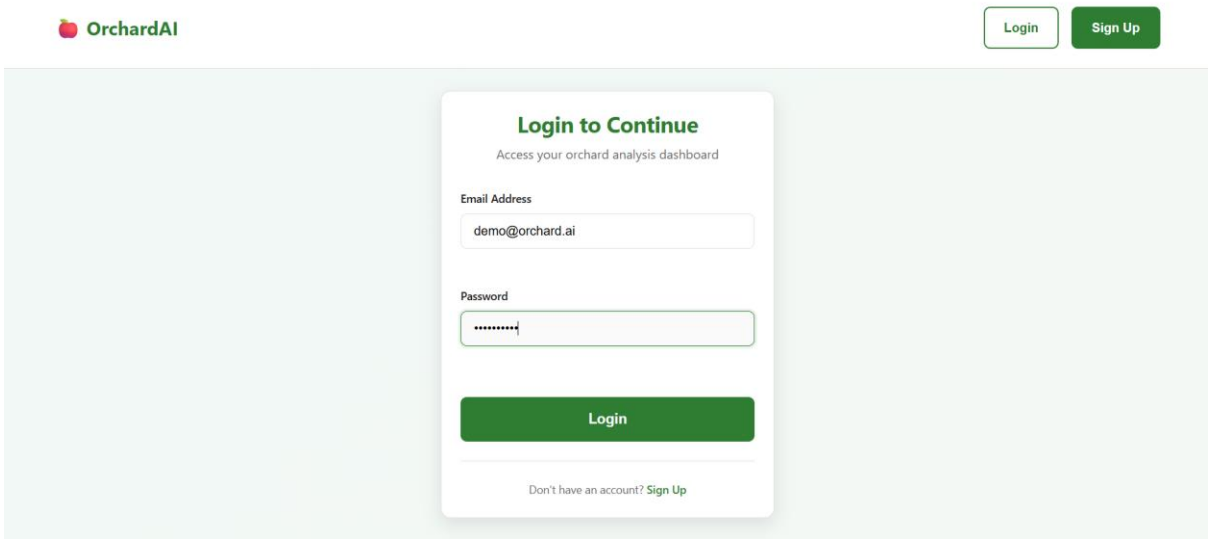

apple_orchard_management
File Edit View Insert Runtime Tools Help
Commands + Code + Text Run all
Files
..
Apple-Scab-Detection-2
apple-powdery_mildew-2
apple_leaf_black_cot-2
apple_leaf_cedar_rust-2
sample_data
[4] 6s
!pip install roboflow
from roboflow import Roboflow
rf = Roboflow(api_key="rIyVizEa3WZ5a61pY6cg")
project = rf.workspace("ayush-shrivastava-jyni9").project("apple-powdery_mildew")
version = project.version(2)
dataset = version.download("yolov8")
Show hidden output
[1] 18s
!pip install roboflow
from roboflow import Roboflow
rf = Roboflow(api_key="TfaJDGwsXtwhN7sHfg9Y")
project = rf.workspace("prashants-workspace-5h9nl").project("apple_leaf_black_cot")
version = project.version(2)
dataset = version.download("yolov8")
Show hidden output
[2] 8s
!pip install roboflow
from roboflow import Roboflow
rf = Roboflow(api_key="TfaJDGwsXtwhN7sHfg9Y")
project = rf.workspace("prashants-workspace-5h9nl").project("apple_leaf_cedar_rust")
version = project.version(2)
dataset = version.download("yolov8")
Show hidden output
[5] 5s
!pip install roboflow
from roboflow import Roboflow
rf = Roboflow(api_key="MRciDJkbwhJjgi0CgnpW")
project = rf.workspace("samars-workspace-6mtle").project("apple-scab-detection-sb3r6")
version = project.version(2)
dataset = version.download("yolov8-obb")
Show hidden output

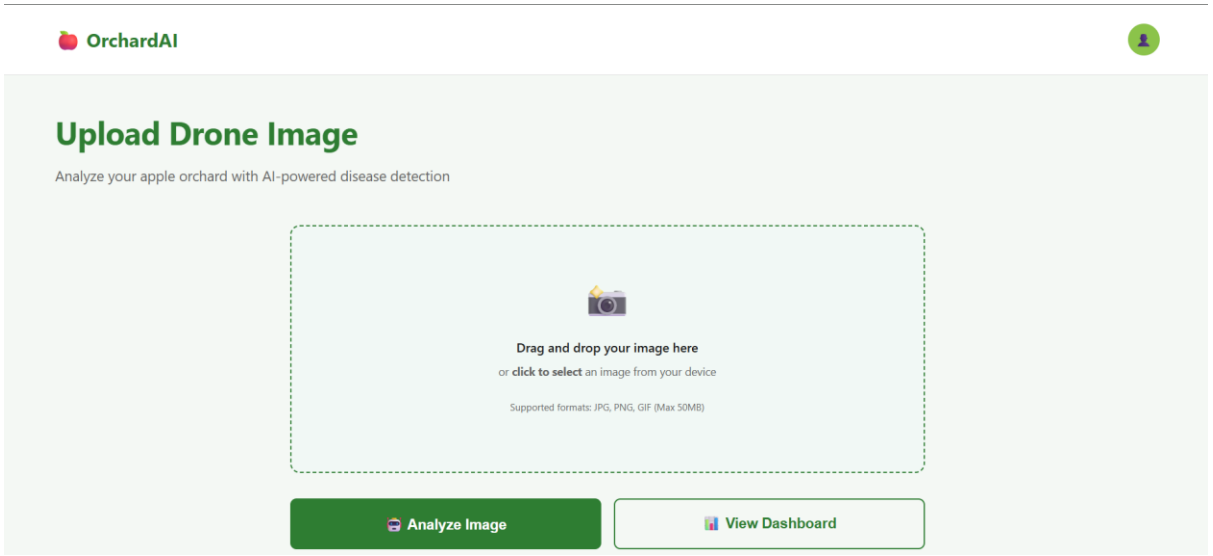
Database:

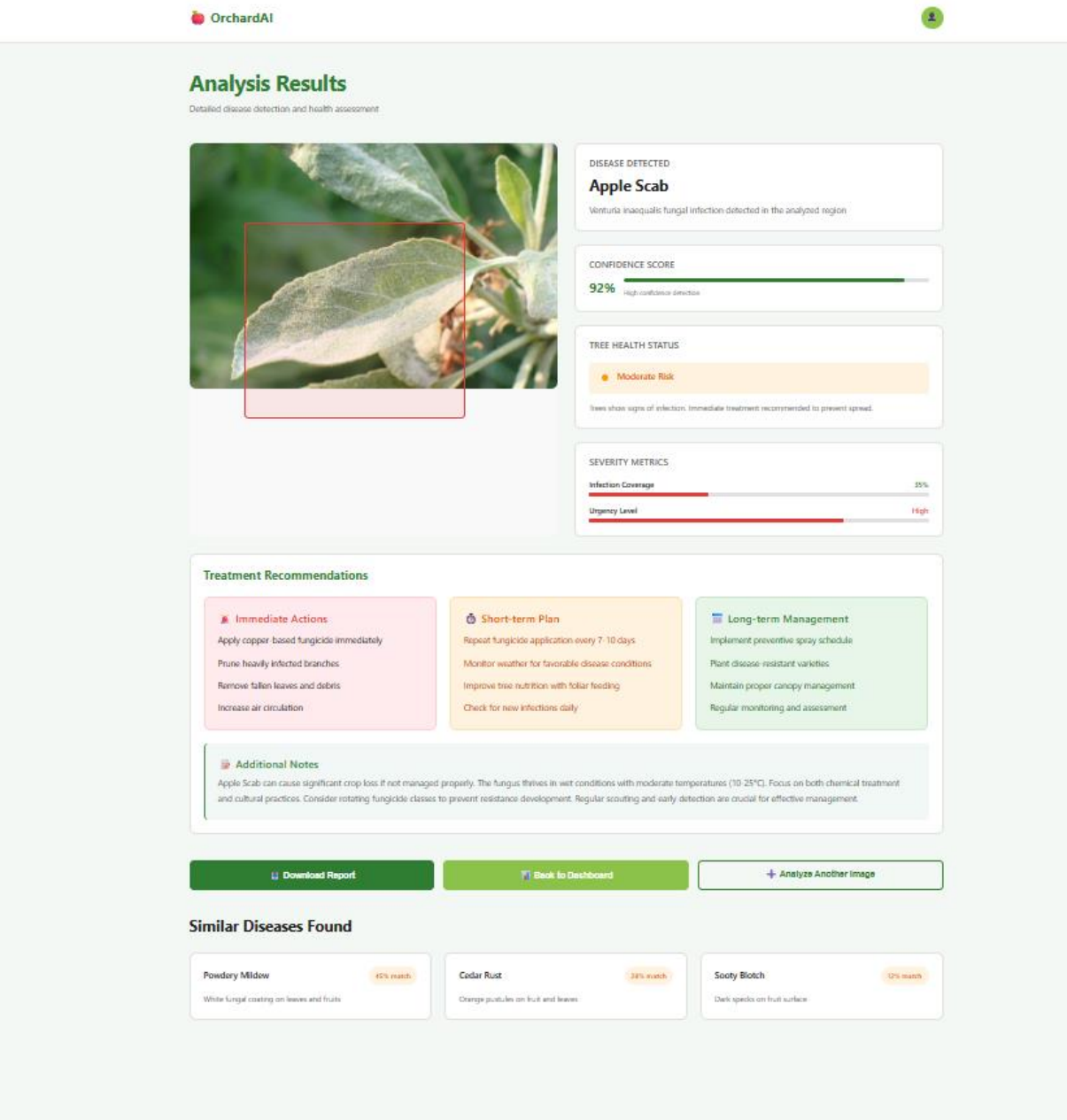
```
Command Prompt - mysql -u

Microsoft Windows [Version 10.0.26200.8037]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Aayush>mysql -u root -p
Enter password: ****
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 11
Server version: 9.2.0 MySQL Community Server - GPL

Copyright (c) 2000, 2025, Oracle and/or its affiliates.

Oracle is a registered trademark of Oracle Corporation and/or its
affiliates. Other names may be trademarks of their respective
owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql>  USE apple_orchard_db;
Database changed
mysql> show Tables;
+----------------------------+
| Tables_in_apple_orchard_db |
+----------------------------+
| advisories                 |
| climate_data               |
| diseases                   |
| feedback                   |
| orchard_health_history     |
| orchard_images             |
| users                      |
| yield_predictions          |
+----------------------------+
8 rows in set (0.13 sec)

mysql> |
```

Frontend:


OrchardAI
Login
Sign Up
AgriTech Innovation
AI-Powered Apple Orchard Monitoring System
Upload orchard images to detect apple diseases using intelligent AI analysis
Upload Drone Image
View Dashboard
92%
Detection Accuracy
500+
Orchards Monitored
24/7
AI Support


OrchardAI
Login
Sign Up
Login to Continue
Access your orchard analysis dashboard
Email Address
demo@orchard.ai
Password
Login
Don't have an account? Sign Up


OrchardAI
Upload Drone Image
Analyze your apple orchard with AI-powered disease detection
Drag and drop your image here
or click to select an image from your device
Supported formats: JPG, PNG, GIF (Max 50MB)
Analyze Image
View Dashboard



3.Technical Execution (DevOps & Git)

A. **Jira Tasks Completed:**

1. Dataset preparation using Roboflow and integration into Google Colab.
2. Preparation of apple disease dataset (4 disease classes).
3. Frontend UI testing and validation.
4. Database schema creation and implementation in MySQL.
5. Initial project structuring for backend integration.

Link://ayushshrivastava.atlassian.net/jira/software/projects/DBISFAOM/boards/2?atlOrigin=eyJpIjoiY2I0Y2FjMmQ2MTliNDBlM2FjYzEyN2U1MjdkY2QzNGMiLCJwIjoiaiJ9

B. **Git Activity:**

1. Updated repository with frontend code and dataset-related work.
2. Maintained version control for ongoing development.
3. Prepared project structure for upcoming backend integration.

Link : https://github.com/yp1818/Drone-Based-Apple-Orchard-Management

C. **CI/CD Status:**

1. Frontend testing completed.
2. ;Database created and validated independently.
3. Backend CI/CD setup is pending and planned for next sprint.

4. Testing Summary

1. Tested frontend UI with multiple image inputs.
2. Verified dataset loading and preprocessing in Google Colab.
3. Validated database tables and relationships in MySQL.
4. Ensured system readiness for integration of frontend, backend, and database.

5. Goals for Next Week

1. Develop backend using Flask with OOP structure and integrate YOLO model (best.pt) for prediction.
2. Create APIs for image upload and disease detection.
3. Connect backend with MySQL database and store prediction results.
4. Integrate frontend with backend to enable complete system flow.
5. Perform end-to-end testing and improve system performance (including adding healthy dataset if possible).

Mentor Signature **Project Guide Signature**